Introduction to Programming

Lecture 29

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Private

Public

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Encapsulation

Data Hiding

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Friend

Friend Function

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Friend Function

```
class Date
{
        --

    friend functionName ( Argument_list ) ;
        --
};
```

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Friend Function

- Friend functions are NOT the members of the class
- The class itself declare it's friend functions
- The prototype of these functions are written inside the class and the key word friend is appended before the function name
- Friend functions have access to private and public members of the class

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Example 1

```
class myClass
{
   friend increment ( myClass , int ) ;
   private:
        int topSecret ;
   public:
        myClass ( )
                { topSecret = 100 ; }
        void Display ( )
                { cout<< "The value of topSecret is "
                        << topSecret ; }
} ;
```

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Example 1

```
void Increment ( myClass A , int i )
{
  A.topSecret += i ;
}
```

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Example 1

```
main ( )
{
  myClass x ;
  x.Display ( ) ;
  Increment ( x , 10 ) ;
  x.Display ( ) ;
}
```

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Example 1: Output

The value of topSecret is 100

The value of topSecret is 110

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Example 2

```
class myClassTwo ;
class myClassOne
{
   private :
            int topSecret ;
   public :
            void Display ( ) ;
            myClassOne ( )
                { topSecret = 100; }
            friend AddBoth ( myClassOne , myClassTwo ) ;
} ;
```

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Example 2

```
class myClassTwo
{
   private:
        int topSecret ;
   public:
        void Display ( ) ;
        myClassTwo ( )
                { topSecret = 200 ; }
        friend AddBoth ( myClassOne , myClassTwo ) ;
} ;
```

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Example 2

```
main ( )
{
  myClassOne A ;
  myClassTwo B ;
  A.Display ( ) ;
  B.Display ( ) ;
  AddBoth ( A , B ) ;
}
```

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Example 2

```
int AddBoth ( myClassOne A , myClassTwo B )
{
    cout << "The value of topSecret in myClassOne object
                is" << A.topSecret ;

    cout << "The value of topSecret in myClassTwo object
                 is"<< B.topSecret ;

    cout << "The sum of topSecret values in "
          << "myClassOne  and myClassTwo object is "
          << A.topSecret + B.topSecret ;
}
```

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Example 3

```
class otherClass ;
class classOne
{
   private :
        int topSecret;
   public :
        void Display ( ) ;
        classOne ( )
                { TopSecret = 100 ; }
        friend otherClass ;
} ;
```

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Straight Line

$$y = mx + c$$

Slope

Intercept on y axis

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Straight Line

```
class straightLine
{
    double slope , intercept ;
    // member function
};
```

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Quadratic

$$y = ax^2 + bx + c$$

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Quadratic

```
class quadratic
{
  private :
     double a , b , c ;
     // member function
};
```

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Limitations

It is NOT Transitive

It is NOT Associative

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